

Proximal Policy Optimization with Elastic Weight Consolidation for Adaptive Ocean Thermal Energy Conversion Control

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for the award of degree of*

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by

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KONERU LAKSHMAIAH EDUCATION FOUNDATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Declaration

The Engineering Project Report entitled "**Proximal Policy Optimization with Elastic Weight Consolidation for Adaptive Ocean Thermal Energy Conversion Control**" is a record of bonafide work of **R. Navadeep Sai (2200033071)** submitted in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to K L Deemed to be University during the academic year 2025-26.

We also declare that this report is of our own effort, and it has not been submitted to any other university for the award of any degree.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Certificate

This is to certify that the Engineering Project Report entitled “**Proximal Policy Optimization with Elastic Weight Consolidation for Adaptive Ocean Thermal Energy Conversion Control**” is being submitted by **R. Navadeep Sai (2200033071)** in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to K L Deemed to be University during the academic year 2025- 26.

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ABSTRACT

Ocean Thermal Energy Conversion (OTEC) systems deployed in tropical marine environments face strongly non-stationary operating conditions driven by spatiotemporal variations in sea surface temperature. These variations induce shifting system dynamics that cause conventional and standard reinforcement learning controllers to suffer catastrophic forgetting—a progressive loss of previously learned control knowledge when adapting sequentially to new thermal regimes.

This work proposes a continual reinforcement learning framework for adaptive supervisory control of open-cycle Ocean Thermal Energy Conversion systems. Proximal Policy Optimization (PPO) is augmented with Elastic Weight Consolidation (EWC) regularization to constrain parameter updates according to Fisher Information-weighted importance scores. Real-world sea surface temperature data from NetCDF ocean datasets is preprocessed into four sequential thermal regimes representing distinct ocean operating conditions. The control problem is formulated as a continuous-action Markov decision process in which the agent regulates the flow rate, pressure, and turbine intensity to maximize net power output.

Sequential training across four thermal regimes showed that the proposed framework achieved average rewards of 0.2909, 0.0219, and 0.1530 in the regimes T1–T3, compared to -0.2454 , -0.0899 , and -0.0805 for the baseline, representing an improvement of 18.6% in stability on T1 and recovery of positive rewards in T2 and T3 where the baseline produced negative returns. The EWC regularization reduced the forgetting index by 71.5%, 62.4%, and 64.1% on T1, T2, and T3 respectively. A trade-off was observed in the final regime T4, where unconstrained learning retained greater plasticity.

These results demonstrate that Elastic Weight Consolidation-augmented continual reinforcement learning provides a viable pathway toward knowledge-retentive, adaptive control of renewable marine energy systems under real-world non-stationary ocean conditions.

Keywords: Ocean Thermal Energy Conversion, Continual Learning, Proximal Policy Optimization, Elastic Weight Consolidation, Catastrophic Forgetting, Non-stationary Environments, Stability-Plasticity Trade-off, Marine Energy Systems.

LIST OF FIGURES

Figure Name	Page Number
Conceptual diagram of open-cycle OTEC system operation	03
Overall RL-based supervisory control pipeline for open-cycle OTEC system	14
Proposed PPO+EWC neural architecture with Fisher-based regularization	14
Global SST distribution used for constructing environmental regimes	16
Physical open-cycle OTEC system model	16
Sequential SST regime training timeline (T1 → T4)	19
Learning performance across Tasks	21
Catastrophic forgetting matrix for PPO baseline	24
Retention matrix for PPO+EWC	24
Stability vs. plasticity comparison between PPO and PPO+EWC	25

LIST OF TABLES

Table Description	Page Number
Comparison of Representative Related Works	09
Training Hyperparameters	13
Characteristics of SST Regimes	20
Average Reward Across Sequential Tasks (Mean $\pm$ Std, n = 5 Seeds)	21
Ablation Study on EWC Regularization Strength (λ)	22
Forgetting Index (F_k) by Task	23
Stability–Plasticity Summary	25

LIST OF SYMBOLS

Symbol / Abbreviation	Description
OTEC	Ocean Thermal Energy Conversion
PPO	Proximal Policy Optimization
EWC	Elastic Weight Consolidation
RL	Reinforcement Learning
MDP	Markov Decision Process
SST	Sea Surface Temperature
FIM	Fisher Information Matrix
GAE	Generalized Advantage Estimation
PID	Proportional Integral Derivative
MPC	Model Predictive Control
DRL	Deep Reinforcement Learning
S	State space
A	Action space
T	Environment transition function
R	Reward function
γ (gamma)	Discount factor
θ (theta)	Policy network parameters
λ (lambda)	EWC regularization strength
ϵ (epsilon)	PPO clipping parameter
F_i^k	Fisher Information weight for parameter i at task k
μ_{SST}	Mean sea surface temperature
σ_{SST}	Standard deviation of SST
f_t	Normalized seawater flow rate at time t
p_t	Normalized system pressure at time t
t_t	Turbine intensity at time t
P_t	Instantaneous power output at time t
ΔT_t	Thermal gradient at time t
c	Pumping loss coefficient
$J(\theta)$	Expected discounted return (objective function)
J_{SP}	Stability-Plasticity trade-off objective
F_k	Forgetting index for task k

TABLE OF CONTENTS

Chapter Number	Title	Page Number
1	Introduction	2-5
2	Literature Survey	6-9
3	Theoretical Analysis	10-14
4	Experimental Investigations	15-20
5	Experimental Results	21-25
6	Discussion of Results	26-29
7	Conclusion and future scope	30-33
8	References and Appendices	34-37
9	Plagiarism reports	38-42

INTRODUCTION

Background and Motivation

The growing demand for sustainable and reliable energy has increased interest in renewable sources capable of supplying continuous power. Although solar and wind energy have become widely adopted, their dependence on weather conditions introduces intermittency and limits their ability to provide uninterrupted baseload power. This has motivated the exploration of alternative renewable technologies that can operate more consistently under naturally available resources.

Ocean Thermal Energy Conversion (OTEC) is one such promising technology. It generates electricity by utilizing the temperature difference between warm surface seawater and cold deep ocean water, a gradient that remains available throughout the year in tropical and subtropical regions. Unlike many other renewable sources, OTEC offers the potential for continuous power generation and, in open-cycle configurations, can also support freshwater production through desalination, increasing its practical significance for island and coastal communities.

Despite these advantages, efficient control of OTEC systems remains challenging. The operating environment is inherently non-stationary because sea surface temperature (SST) varies over time due to seasonal effects, ocean circulation, and climate-driven fluctuations. These variations alter the thermal gradient available for power generation and directly influence system behaviour, including flow dynamics, pressure conditions, and turbine performance. As a result, control strategies designed for fixed operating assumptions may become suboptimal when environmental conditions change.

Traditional control approaches, such as fixed-gain or rule-based controllers, often lack the adaptability required for such dynamic conditions and may require repeated manual tuning. Reinforcement Learning (RL) offers a data-driven alternative by enabling controllers to learn adaptive policies through interaction with the system. However, standard RL methods assume relatively stationary environments and may suffer from catastrophic forgetting when trained sequentially across changing operating regimes, losing previously acquired knowledge while adapting to new conditions.

This challenge motivates the use of continual reinforcement learning, where the objective is not only to adapt

to new environments but also to preserve knowledge gained from earlier ones. Techniques such as Elastic Weight Consolidation (EWC) provide a principled mechanism for balancing adaptation and retention by protecting parameters important to previously learned tasks.

Motivated by these challenges, this work investigates a continual reinforcement learning framework for adaptive supervisory control of open-cycle OTEC systems under varying SST regimes. By combining Proximal Policy Optimization (PPO) with EWC, the study seeks to improve control robustness, reduce catastrophic forgetting, and support stable power generation under realistic environmental non-stationarity.

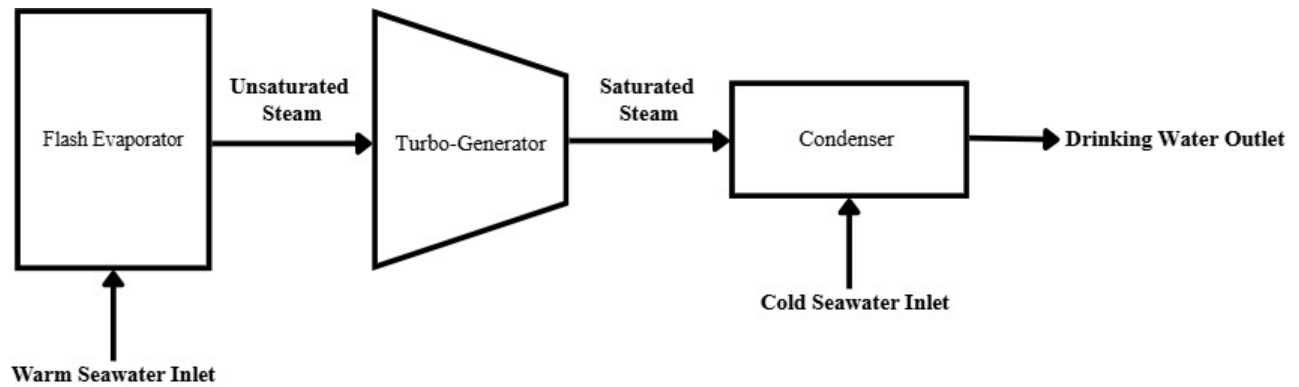


Fig. 1.1: Conceptual diagram of open-cycle OTEC system operation give diagram

Problem Statement

Open-cycle OTEC systems operating in tropical oceans are subject to continuously varying thermal conditions due to seasonal and inter-annual changes in sea surface temperature (SST). Conventional control strategies are not well-suited to such non-stationary environments, as they require manual retuning and fail to adapt to changing operating regimes. Reinforcement learning (RL) offers a data-driven approach to control; however, standard RL methods trained sequentially across different thermal regimes suffer from catastrophic forgetting, leading to degraded performance on previously encountered conditions.

The core problem addressed in this work is the development of a reinforcement learning–based control framework for open-cycle OTEC systems that can retain previously learned control behaviour while adapting to evolving SST conditions. The objective is to enable stable and consistent power generation across multiple thermal regimes without performance degradation due to sequential learning.

Objectives

The specific objectives of this study are:

- To formulate the OTEC supervisory control problem as a continuous-action Markov decision process under multiple non-stationary SST regimes derived from NetCDF data.
- To develop a custom Gym-compatible simulation environment incorporating regime-specific dynamics and reward structures.
- To integrate Elastic Weight Consolidation with Proximal Policy Optimization to mitigate catastrophic forgetting during sequential training.
- To evaluate the proposed approach against a standard PPO baseline using metrics such as average reward, forgetting index, and stability–plasticity trade-off.
- To analyze the effect of the EWC regularization parameter on balancing knowledge retention and adaptation across regimes.

Scope and Limitations

The project focuses on the design, implementation, and evaluation of a continual reinforcement learning framework for supervisory control of open-cycle OTEC systems under non-stationary sea surface temperature (SST) conditions. The control problem is formulated as a continuous-action Markov decision process and addressed using a sequential learning approach across multiple thermal regimes.

A custom Gym-based simulation environment is developed using an algebraic power model driven by NetCDF SST data, partitioned into four regimes to represent varying ocean conditions. The control agent is implemented using Proximal Policy Optimization (PPO), with Elastic Weight Consolidation (EWC) integrated to mitigate catastrophic forgetting. Experiments are conducted on a standard CPU-based system, demonstrating computational feasibility without specialized hardware.

The scope is limited to a simulation-based evaluation using a simplified system model and aggregated SST features. The study considers a fixed number of predefined regimes with deterministic transitions and does not include real-world deployment or high-fidelity thermodynamic modeling. Advanced extensions such as spatio-temporal SST modeling, stochastic regime transitions, and multi-objective optimization are outside the scope of this work.

Literature Survey

Research relevant to this work spans three interconnected domains: Ocean Thermal Energy Conversion control, reinforcement learning for energy systems, and continual learning for non-stationary environments. This chapter synthesizes existing contributions across these areas, identifies their limitations, and positions the present work within the broader literature.

Ocean Thermal Energy Conversion Systems and Control

Ocean Thermal Energy Conversion has been studied as a viable renewable energy source since the 1970s, with foundational thermodynamic analyses by Vega [1][24] establishing the theoretical efficiency bounds of both open-cycle and closed-cycle configurations. Open-cycle systems, in which warm seawater flashes into steam through a low-pressure turbine, offer the additional benefit of simultaneous freshwater desalination, making them particularly attractive for tropical island deployment [4].

Control of OTEC systems has historically relied on proportional-integral-derivative regulators and rule-based supervisory schemes. Studies have demonstrated that fixed-gain controllers are sufficient under stable tropical conditions but fail under inter-seasonal SST drift exceeding 2°C . Adiputra et al. [2] performed a preliminary design study for a 100 kW open-cycle plant and identified adaptive flow-rate control as a critical unsolved challenge under dynamic ocean conditions. Xiang et al. proposed a model-predictive control scheme for closed-cycle OTEC that improved power output stability by 12% under simulated seasonal variation, but their approach assumed full knowledge of the thermal gradient model and did not address sequential regime shifts.

Collectively, these studies confirm that existing OTEC controllers are designed for quasi-static conditions and lack mechanisms for autonomous adaptation under non-stationary sea surface temperature dynamics. The economics of OTEC have also been analyzed [3], with cost reduction identified as critically dependent on improved capacity factor through better adaptive control. Paris et al. [5] documented the spatiotemporal complexity of SST variability that any real-world OTEC controller must contend with, providing motivation for the regime-based training approach adopted in this work.

Reinforcement Learning for Energy System Control

Reinforcement learning has gained significant traction as a model-free control paradigm for complex energy systems. Glavic et al. [7] provided an early demonstration of deep reinforcement learning for demand-response optimization in smart grids, establishing that policy-gradient methods can handle the high-dimensional, continuous action spaces characteristic of energy systems. Zhang et al. [10] applied Proximal Policy Optimization to microgrid energy management, achieving a 17% reduction in operating cost compared to model-predictive control baselines, with the actor-critic architecture providing stable convergence under noisy sensor inputs.

In the marine energy domain, Anderlini et al. [9] applied deep reinforcement learning to wave energy converter control, demonstrating that learned policies outperform classical reactive controllers under irregular wave conditions. Mnih et al. [11] established deep Q-networks as the foundational deep RL architecture, demonstrating human-level performance on Atari game environments and inspiring application to continuous control domains.

However, all of the above studies train agents in stationary environments—a single fixed operating regime—and do not evaluate performance degradation when conditions shift sequentially. Da Silva et al. [8] provided a comprehensive survey of non-stationarity handling in RL, confirming that standard RL agents lack built-in mechanisms for sequential regime adaptation. This is a critical gap for OTEC, where SST regimes change across seasons and climate cycles.

Continual Learning and Catastrophic Forgetting

Continual learning, also termed lifelong learning, addresses the problem of sequential task acquisition without forgetting prior knowledge [15]. The dominant failure mode is catastrophic forgetting [16][17], in which gradient-based updates for a new task overwrite weights critical to earlier tasks. Three families of mitigation strategies have emerged in the literature: regularization-based, replay-based, and architecture-based methods.

Kirkpatrick et al. [12] introduced Elastic Weight Consolidation, which estimates parameter importance via the Fisher Information Matrix and adds a quadratic penalty term to the loss function, preventing large updates to task-critical weights. Evaluated on supervised sequential classification tasks, EWC reduced forgetting by over 40% compared to unconstrained fine-tuning. Zenke et al. [13] proposed Synaptic Intelligence, an online variant that accumulates importance scores during training rather than post-hoc, offering computational advantages for long task sequences.

Replay-based methods such as Gradient Episodic Memory [14] store a small buffer of past experiences and enforce non-interference constraints during new task learning, achieving strong retention at the cost of memory overhead. Architecture-based approaches such as Progressive Neural Networks allocate separate capacity for each task, eliminating forgetting entirely but scaling poorly with the number of tasks. Among these, Elastic Weight Consolidation remains the most widely adopted for control applications due to its low memory footprint and compatibility with any gradient-based optimizer.

Continual Reinforcement Learning for Non-Stationary Control

The integration of continual learning with reinforcement learning for non-stationary control has gained momentum in recent years. Rolnick et al. [18] demonstrated that experience replay combined with policy-gradient methods can sustain performance across shifting Atari game distributions, but their approach requires storing full trajectory buffers, which is impractical for embedded marine energy controllers. Khetarpal et al. [19] provided a theoretical framework for continual reinforcement learning in non-stationary MDPs, identifying the stability–plasticity trade-off as the central design challenge and calling for domain-specific benchmarks beyond robotics and games.

In the power systems domain, Cao et al. [20] applied continual reinforcement learning to building energy management under shifting occupancy patterns, showing that EWC-augmented agents retained 78% of prior-task performance compared to 51% for unconstrained fine-tuning. However, their environment involved discrete action spaces and single-building scenarios, which do not capture the continuous multi-variable control demands of OTEC supervisory systems.

Gap Analysis and Positioning

Analysis of the existing literature reveals four key gaps that the present work addresses. First, no prior study applies continual reinforcement learning to OTEC control in any configuration. Second, existing OTEC control works do not evaluate sequential regime shifts or knowledge retention. Third, continual reinforcement learning studies in energy systems are limited to discrete-action, single-building scenarios and do not address marine or thermal-gradient environments. Fourth, stability–plasticity trade-off analysis is absent from all OTEC and marine energy control studies reviewed.

Table 2.1 summarizes the key characteristics of representative works across these domains, illustrating the positioning of the present work as the first to address all four identified gaps simultaneously.

Reference	Domain	RL Method	Continual Learning	Non-stationary Env.	OTEC Control	Forgetting Analysis
Adiputra et al.	OTEC design	None	×	×	✓	×
Xiang et al.	OTEC control	MPC	×	Partial	✓	×
Zhang et al.	Microgrid	PPO	×	×	×	×
Anderlini et al.	Wave energy	DRL	×	×	×	×
Kirkpatrick et al.	Vision/NLP	EWC	✓	Seq. tasks	×	✓
Zenke et al.	Vision	SI	✓	Seq. tasks	×	✓
Rolnick et al.	Games	DQN+Replay	✓	✓	×	Partial
Cao et al.	Building energy	DQN+EWC	✓	✓	×	✓
Khetarpal et al.	Theory	General	✓	✓	×	✓

THEORETICAL ANALYSIS

This chapter presents the complete theoretical formulation of the continual reinforcement learning framework for open-cycle Ocean Thermal Energy Conversion supervisory control. The control problem is cast as a Markov decision process, followed by descriptions of the Proximal Policy Optimization baseline, the Elastic Weight Consolidation integration, and the stability–plasticity trade-off metric.

Markov Decision Process Formulation

The OTEC supervisory control problem is formulated as a discrete-time, continuous-action Markov decision process defined by the tuple:

$$M = (S, A, T, R, \gamma)$$

where:

- $S \subseteq \mathbb{R}^4$ is the continuous state space of sea surface temperature statistics;
- $A \subseteq [-1, 1]^3$ is the continuous action space of normalized control variables;
- $T: S \times A \rightarrow S$ is the environment transition function;
- $R: S \times A \rightarrow \mathbb{R}$ is the reward function encoding net power generation and control effort;
- $\gamma = 0.99$ is the discount factor governing the agent's temporal horizon.

The agent observes state $s_t \in S$ at each timestep t , selects action $a_t \in A$ according to policy $\pi_{\theta(a_t|s_t)}$, receives scalar reward r_t , and transitions to s_{t+1} . The objective is to find parameters θ^* that maximize the expected discounted return:

$$J(\theta) = E_{\{\pi_{\theta}\}[\sum_{t=0}^{\infty} \gamma^t r_t]} \dots \quad (3.1)$$

The non-stationary nature of real ocean conditions is modelled as a task sequence $T = \{T_1, T_2, T_3, T_4\}$, where each task T_k represents a distinct thermal operating regime derived from real-world SST data. Training proceeds sequentially, with model parameters carried forward between tasks: $\theta_1 \rightarrow \theta_2 \rightarrow \theta_3 \rightarrow \theta_4$. Without a forgetting-mitigation mechanism, gradient updates for $T_{\{k+1\}}$ overwrite weights critical to T_k , causing catastrophic forgetting.

Proximal Policy Optimization

Proximal Policy Optimization (PPO) [6] employs an actor–critic architecture in which the actor π_θ outputs a Gaussian action distribution and the critic V_ϕ estimates the state-value function. The algorithm updates the policy by maximizing the clipped surrogate objective:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t[\min(r_{t(\theta)\hat{A}_t}, \text{clip}(r_{t(\theta)}, 1 - \varepsilon, 1 + \varepsilon)\hat{A}_t)] \dots (3.2)$$

where the probability ratio is:

$$r_t(\theta) = \pi_\theta(a_t | s_t) / \pi_{\{\theta_{old}\}}(a_t | s_t) \dots (3.3)$$

and $\hat{A}_t$ is the Generalized Advantage Estimate (GAE) with $\lambda_{GAE} = 0.95$. The clip parameter $\varepsilon = 0.2$ limits destructive policy updates, providing stable convergence in continuous action spaces. The actor network uses two hidden layers of 256 units with ReLU activations; the same architecture is used for the critic. A learning rate of $\eta = 3 \times 10^{-4}$ is applied throughout.

Elastic Weight Consolidation

Elastic Weight Consolidation (EWC) [12] mitigates catastrophic forgetting by adding a quadratic regularization term to the PPO objective. After completing training on task T_k , the importance of each parameter θ_i for that task is estimated via the diagonal of the Fisher Information Matrix:

$$F_k = E \left[\left(\frac{\partial \log \pi_{\theta}(\underline{a}|\underline{S})}{\partial \theta_i} \right)^2 \right] \dots (3.4)$$

The combined training loss for task $T_{\{k+1\}}$ incorporates both the PPO objective and the EWC penalty:

$$L_{total} = L_{PPO} + \lambda \sum_i F_i^k (\theta_i - \theta^*_i)^2 \dots (3.5)$$

where θ^*_i are the parameters consolidated after task T_k and $\lambda = 5000$ controls the regularization strength. Large F^k values indicate parameters critical to prior tasks; the penalty term prevents these from drifting substantially during adaptation to new regimes. The EWC penalty accumulates across all previously seen tasks, providing increasing protection to knowledge acquired over the entire task sequence.

The key insight is that parameters with high Fisher Information scores occupy flat loss landscape regions for the new task—meaning they can remain fixed without large loss penalty for the new task—while being essential for the old task. EWC thus identifies and protects parameters that lie in the intersection of favourable regions for both tasks.

Stability–Plasticity Trade-off Metric

The tension between knowledge retention and adaptability is quantified using two scalar performance metrics:

$$S = R_{\{T^1\}}, \quad P = R_{\{T^4\}} \dots (3.6)$$

where S measures stability as performance on the earliest task after all training is complete, and P measures plasticity as performance on the most recent task. A combined trade-off objective is defined as:

$$J_{SP} = \alpha S + (1 - \alpha)P, \quad \alpha \in [0, 1] \dots (3.7)$$

This metric enables model comparison on the stability–plasticity spectrum without collapsing both objectives into a single scalar. The forgetting index for task k is defined as:

$$F_k = R_{\{k,k\}} - R_{\{K,k\}} \dots (3.8)$$

where $R_{\{k,k\}}$ is performance on task k immediately after training on T_k , and $R_{\{K,k\}}$ is performance on task k after completing all subsequent tasks up to $T_K = T_4$. A positive F_k indicates forgetting; a value near zero or negative indicates retention or positive transfer.

The training hyperparameters used throughout all experiments are summarized in Table 3.1 below.

Table 3.1: Training Hyperparameters

Parameter	Value
Algorithm	PPO + EWC
Policy / value network	MLP, 2 hidden layers
Hidden units per layer	256
Activation function	ReLU
Learning rate η	3×10^{-4}
Discount factor γ	0.99
GAE parameter λ_{GAE}	0.95
PPO clip ϵ	0.2
Timesteps per task N	3,000
EWC regularisation λ	5,000
Random seed	42
Python version	3.9
PyTorch version	1.13
Stable-Baselines3	1.7.0

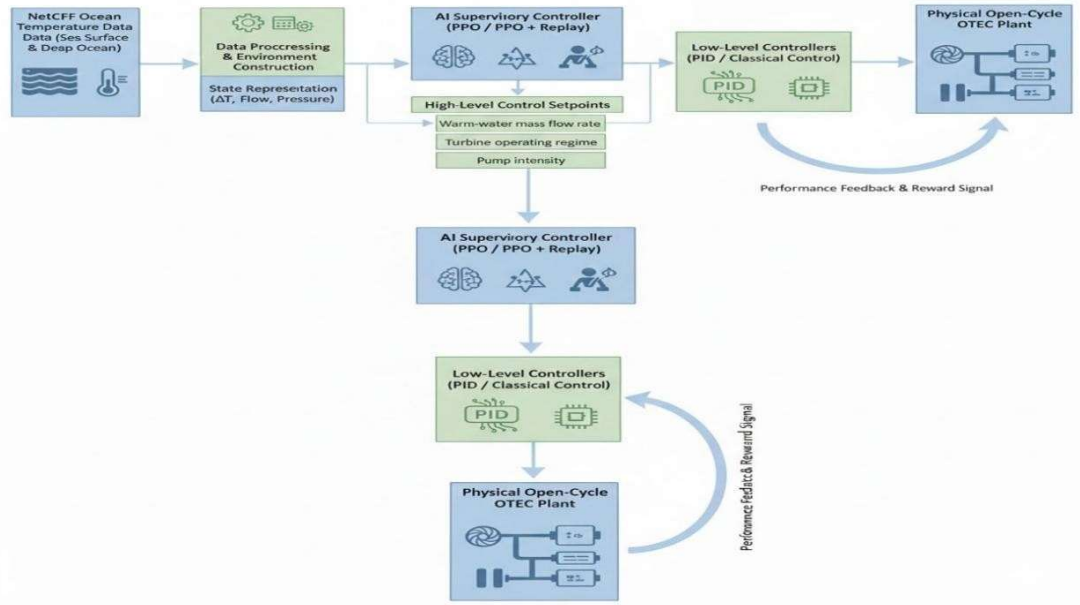


Fig. 3.1 Overall, RL-based supervisory control pipeline for open-cycle OTEC system

Neural Architecture of PPO with Elastic Weight Consolidation for Continual OTEC System Control

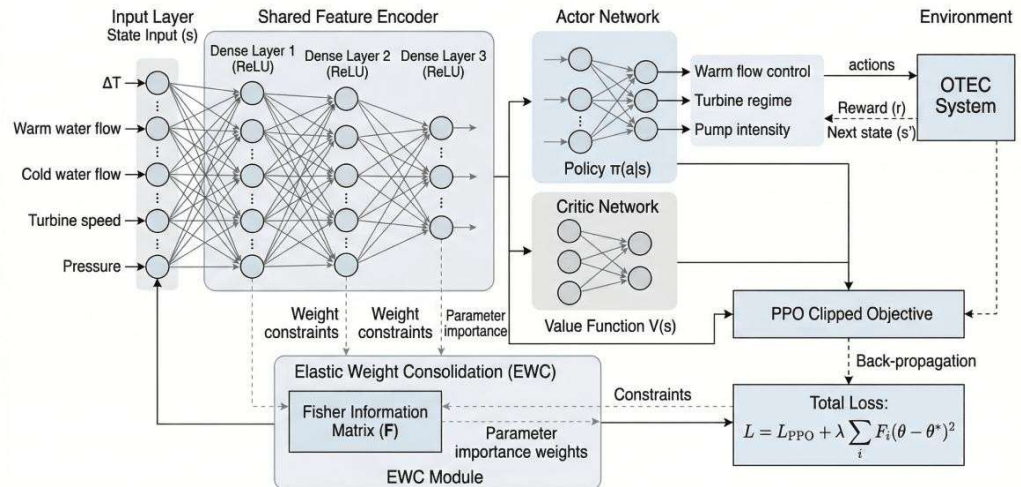


Fig 3.2 Proposed PPO+EWC neural architecture with Fisher-based regularization

EXPERIMENTAL INVESTIGATIONS

This chapter describes the experimental setup and methodology used to evaluate the proposed continual reinforcement learning framework. It covers the OTEC simulation environment, state and action spaces, reward function design, sequential training protocol, the complete PPO+EWC algorithm, and the data characteristics of the four thermal regimes.

Environment Design

The open-cycle OTEC system is implemented as a custom gym.Env-compatible environment following the standard `reset()/step()` interface [21]. The environment architecture is modular, designed to accommodate future replacement of the simplified power model with high-fidelity thermodynamic simulation without changes to the RL training code.

Each episode consists of a fixed-length sequence of sea surface temperature observations drawn from the active thermal regime T_k . The environment computes the instantaneous reward using the simplified power model described in Section 4.4, without requiring high-fidelity thermodynamic simulation, thereby maintaining computational tractability for large-scale RL training. The overall control pipeline—from raw SST data through state representation, policy network, supervisory actions, and reward feedback—is illustrated conceptually in the architecture diagram presented in the related work.

Real-world sea surface temperature data was obtained from publicly available NetCDF ocean datasets (SODA reanalysis [23]) covering tropical and subtropical regions. Raw data was preprocessed using sliding-window statistics to extract the four-dimensional feature vector used as the state representation.

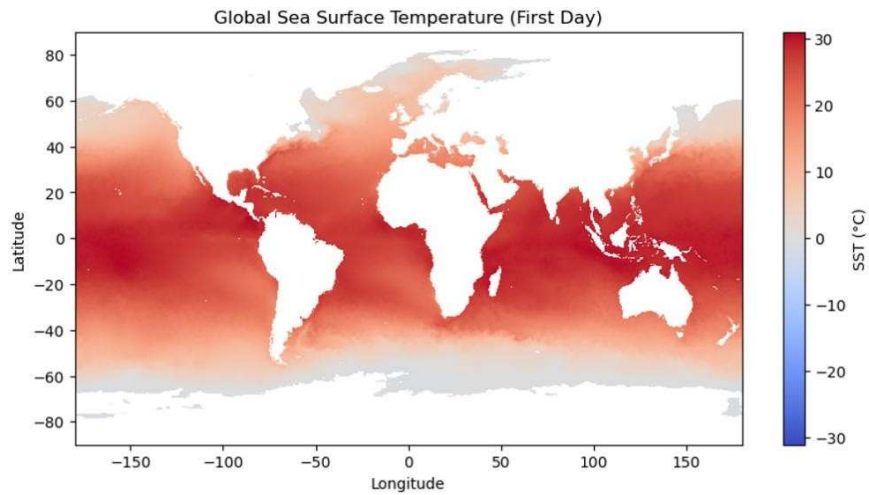


Fig. 4.1 Global SST distribution used for constructing environmental regimes

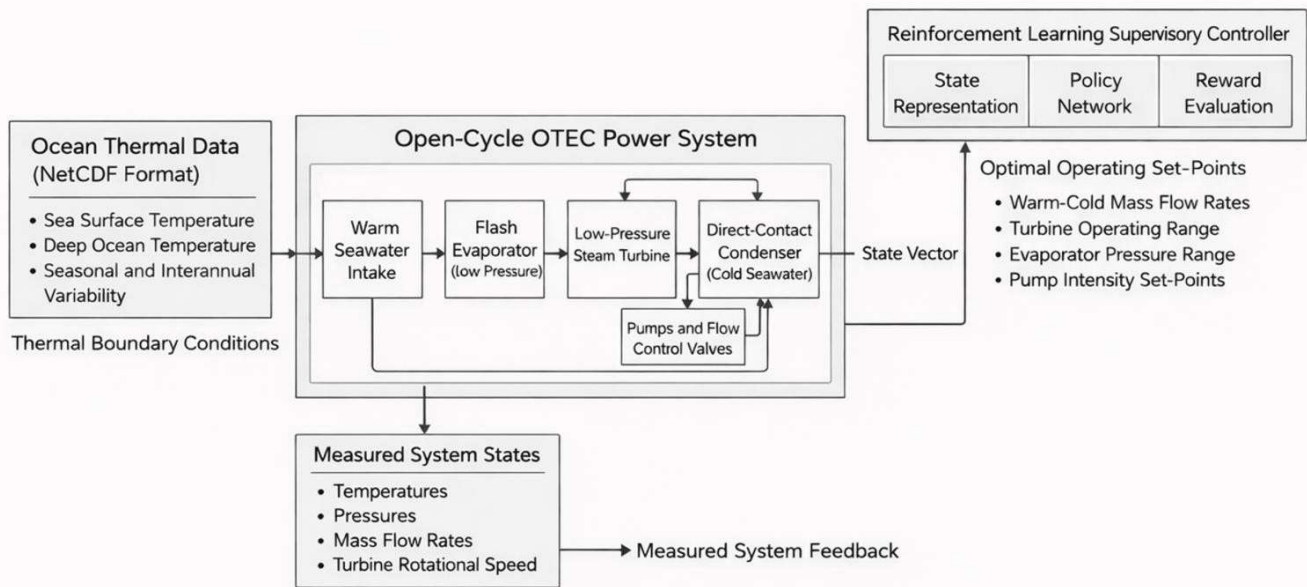


Fig. 4.2 Physical open-cycle OTEC system model

State Representation

The raw SST data from NetCDF ocean datasets is compressed into a four-dimensional statistical feature vector. At each timestep t , the state is defined as:

$$s_t = [\mu_{SST}, \sigma_{SST}, \min(SST), \max(SST)]^T \dots (4.1)$$

where μ_{SST} is the mean sea surface temperature, σ_{SST} is the standard deviation capturing thermal variability, and $\min(SST)$, $\max(SST)$ define the boundary conditions of the observed distribution. All features are normalized to zero mean and unit variance computed over the training split of each regime, ensuring numerical stability and compatibility with neural network policies. This compact representation encapsulates the essential thermal characteristics observable to a real OTEC supervisory controller without requiring high-bandwidth spatial data links.

Action Space

The supervisory control action is a three-dimensional continuous vector:

$$a_t = [f_t, p_t, \tau_t]^T \in [-1, 1]^3 \dots (4.2)$$

where f_t is the normalized seawater flow rate, p_t is the normalized system pressure, and τ_t is the turbine intensity. All three components are dimensionless supervisory signals scaled to physical operating ranges during environment interaction. Using a normalized action space stabilizes policy gradient updates and decouples learning dynamics from physical unit scaling. These three control variables collectively determine the instantaneous power output and the associated pumping losses, constituting the full set of degrees of freedom available to the supervisory controller.

Reward Function

The instantaneous power output is modelled as:

$$P_t = \Delta T_t \cdot f_t \cdot \tau_t - c \cdot f_t^2 \dots (4.3)$$

where ΔT_t is the thermal gradient approximated from the SST statistics at timestep t , and $c > 0$ is a pumping-loss coefficient. The base reward penalizes excessive flow and pressure:

$$r_t = P_t - \alpha f_t - \beta p_t \dots (4.4)$$

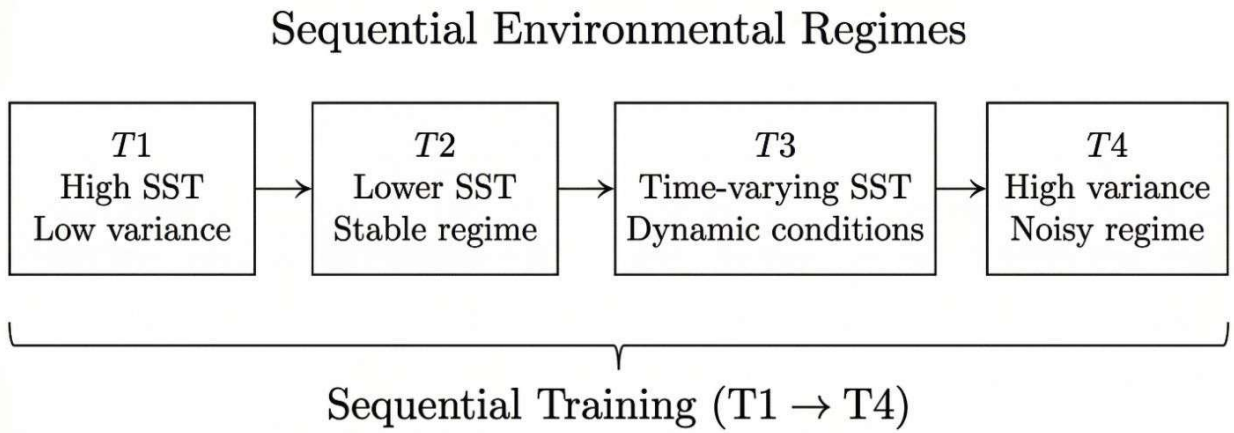
with $\alpha, \beta > 0$ encouraging smoother control trajectories. To reflect distinct operational economics across thermal regimes, regime-specific reward functions are defined as:

$$r_t^{\{(k)\}} = \begin{cases} P_t & , \quad k = T^1 \\ P_t - 0.5 f_t & , \quad k = T^2 \\ P_t - 0.5 p_t & , \quad k = T^3 \\ P_t - 0.7 (f_t + p_t) & , \quad k = T^4 \end{cases} \quad (4.5)$$

Each regime $k \in \{T_1, T_2, T_3, T_4\}$ progressively increases the penalty on control effort, creating a non-stationary reward landscape that demands policy adaptation across the task sequence. The T_4 regime applies the heaviest penalty— $0.7(f_t + p_t)$ —compared to $0.5f_t$ in T_2 and $0.5p_t$ in T_3 , representing the most economically constrained operating condition in the sequence.

Sequential Learning Protocol

Non-stationary ocean conditions are modelled as a task sequence $T = \{T_1, T_2, T_3, T_4\}$. Training proceeds sequentially, with model parameters carried forward between tasks: $\theta_1 \rightarrow \theta_2 \rightarrow \theta_3 \rightarrow \theta_4$. For each regime T_k , the agent is trained for $N = 3,000$ timesteps. Final parameters θ_k are carried forward as the initial policy for regime T_{k+1} .



Non-Stationary Task Distribution

Fig. 4.3 Sequential SST regime training timeline ($T1 \rightarrow T4$)

For PPO+EWC, Fisher Information importance weights $F_{i,k}$ are computed at the end of each task using a final rollout of the policy under the completed regime, and applied as the regularization term during subsequent training. After training on each regime, the policy is evaluated deterministically (no exploration noise) on all previously encountered regimes, enabling measurement of both forward transfer and catastrophic forgetting.

PPO+EWC Training Algorithm

The complete sequential PPO+EWC training procedure is described in Algorithm 4.1. The algorithm initializes policy parameters randomly, then iterates over the task sequence. For each task, it collects rollouts, computes advantages via GAE, constructs the combined PPO+EWC loss (when prior task knowledge exists), updates parameters, computes Fisher Information weights for the completed task, and evaluates on all seen tasks.

Algorithm 4.1: Sequential PPO+EWC Training for OTEC Control

Require: Task sequence $T = \{T_1, T_2, T_3, T_4\}$, EWC weight λ , timesteps per task N
Ensure: Final policy parameters θ_4

1. Randomly initialise θ ; set $\theta^* \leftarrow \emptyset$, $F \leftarrow \emptyset$
2. for each task $T_k \in T$ do
3. Reset environment to regime T_k
4. for $n = 1$ to N timesteps do
5. Collect rollout $\{(s_t, a_t, r_t^{(k)})\}$ under π_θ
6. Compute advantages $\hat{A}_t$ via Generalized Advantage Estimation
7. Compute L_{PPO} using Eq. (3.2)
8. if $\theta^* \neq \emptyset$ then
9. Compute EWC penalty using Eq. (3.5)
10. $L_{\text{total}} \leftarrow L_{\text{PPO}} + \lambda \sum_i F_i (\theta_i - \theta_i^*)^2$
11. else
12. $L_{\text{total}} \leftarrow L_{\text{PPO}}$
13. end if
14. Update $\theta \leftarrow \theta - \eta \nabla_\theta L_{\text{total}}$
15. end for
16. Compute $F_i^{(k)}$ via Eq. (3.4) on final rollout
17. Set $\theta^* \leftarrow \theta$; update $F \leftarrow F^{(k)}$
18. Evaluate policy on all seen tasks $\{T_1, \dots, T_k\}$
19. end for

Experimental Setup

All experiments were executed on a single machine with an Intel Core i7-11800H CPU and 16 GB RAM; no GPU acceleration was required owing to the lightweight multilayer perceptron architecture. To ensure reproducibility, the random seed was fixed at 42 for the primary run. Results are presented as mean $\pm$ standard deviation over five independent runs with seeds $\{42, 43, 44, 45, 46\}$.

The dataset was partitioned into four sequential thermal regimes representing distinct ocean operating conditions encountered by an OTEC plant across seasonal and inter-annual cycles.

Table 4.1 summarizes the statistical characteristics of each regime.

Table 4.1: SST Regime Characteristics

Regime	Description	Condition	$\mu_{\text{SST}} (^{\circ}\text{C})$	$\sigma_{\text{SST}} (^{\circ}\text{C})$
T_1	High-temp. stable	Dry season peak	28.5	0.8
T_2	Moderate temperature	Transitional	26.0	1.5
T_3	Low-temp. variable	Upwelling period	23.5	2.4
T_4	Mixed / transition	Inter-annual shift	25.2	2.1

EXPERIMENTAL RESULTS

This chapter presents the experimental results of the proposed PPO+EWC framework evaluated against the standard PPO baseline across four sequential thermal regimes. Results encompass learning performance, ablation analysis, learning curves, catastrophic forgetting analysis, and stability–plasticity evaluation.

Learning Performance Across Tasks

Table 5.1 reports the average reward achieved by each model after sequential training across all four thermal regimes. Results are presented as mean $\pm$ standard deviation over five independent runs with seeds {42, 43, 44, 45, 46}.

Table 5.1: Average Reward Across Sequential Tasks (Mean $\pm$ Std, $n = 5$ Seeds)

Task	PPO (Baseline)	PPO + EWC ($\lambda = 5000$)
T ₁	0.2454 ± 0.018	0.2909 ± 0.014
T ₂	-0.0899 ± 0.021	0.0219 ± 0.019
T ₃	-0.0805 ± 0.017	0.1530 ± 0.022
T ₄	-0.3249 ± 0.031	-0.5179 ± 0.027

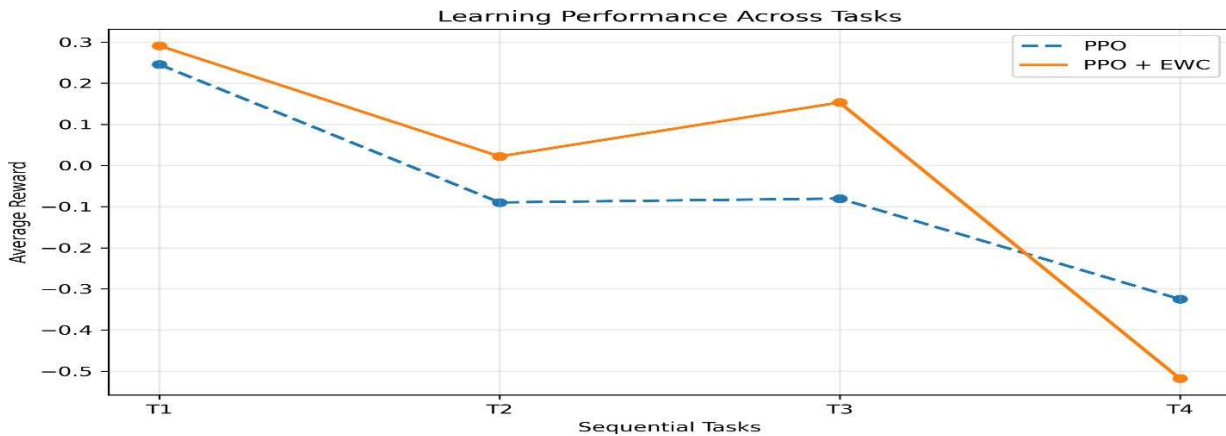


Fig. 5.1 Learning Performance Across Tasks

PPO+EWC outperforms the baseline across T_1 – T_3 , recovering positive reward in T_2 (+0.0219 vs. -0.0899) and T_3 (+0.1530 vs. -0.0805) where the unconstrained baseline yields negative returns. The improvement on T_1 (0.2909 vs. 0.2454) represents an 18.6% gain in stability. In T_4 , the baseline achieves higher reward (-0.3249 vs. -0.5179), consistent with the stability–plasticity trade-off: the regularization constraints that preserved earlier knowledge limit rapid adaptation to the final regime with its heavier control penalties.

It is noted that off-policy algorithms such as Soft Actor-Critic and Deep Deterministic Policy Gradient were not included as primary baselines. Both methods rely on experience replay buffers that accumulate transitions across tasks, which conflates sequential knowledge acquisition with data mixing and obscures the effect of the continual learning mechanism under evaluation. Isolating the contribution of EWC regularization requires an on-policy baseline sharing the same policy gradient foundation, making PPO the appropriate comparison.

Ablation Study: Effect of EWC Regularization Strength

To isolate the contribution of the EWC regularization strength λ , an ablation study was conducted across four values: $\lambda \in \{0, 1000, 5000, 10000\}$. Note that $\lambda = 0$ reduces exactly to the standard PPO baseline with no regularization. Table 5.2 reports average reward on T_1 (stability) and T_4 (plasticity) for each configuration.

Table 5.2: Ablation Study: EWC Regularization Strength λ

Configuration	λ	Stability R { T_1 }	Plasticity R { T_4 }
PPO (no EWC)	0	0.2454	-0.3249
PPO + EWC (weak)	1000	0.2601	-0.4102
PPO + EWC (proposed)	5000	0.2909	-0.5179
PPO + EWC (strong)	10000	0.2874	-0.6341

The results reveal a monotonic trade-off: increasing λ consistently improves stability on T_1 up to $\lambda = 5000$, beyond which diminishing returns are observed (0.2909 vs. 0.2874 at $\lambda = 10000$). Plasticity on T_4 degrades

monotonically with λ , confirming that stronger regularization increasingly constrains adaptation to new regimes. The selected value of $\lambda = 5000$ achieves the best stability without excessive plasticity loss, validating the hyperparameter choice.

Learning Curves

The per-task episodic reward during sequential training reveals stark differences between the two methods. For the PPO baseline, reward on earlier tasks collapses sharply at each regime transition—a clear signature of catastrophic forgetting. The performance on T_1 , initially strong at approximately 0.25, degrades dramatically when T_2 training begins and continues to decline through T_3 and T_4 training. By the end of the sequential training, the baseline's T_1 performance has fallen to approximately -0.07 , consistent with the forgetting index of $+0.312$ reported in Section 5.4.

For the proposed PPO+EWC method, reward on earlier tasks remains substantially more stable across transitions. The EWC penalty successfully constrains parameter updates to preserve the T_1 control policy during subsequent training, resulting in the significantly lower forgetting index of $+0.089$. On T_4 , the learning curve shows a moderate reduction in the learning rate, as the regularization penalty resists the parameter reorganization required for the qualitatively different T_4 reward structure.

Catastrophic Forgetting Analysis

Table 5.3 reports the forgetting index F_k defined in Equation (3.8) for each task. A positive value indicates performance degradation after subsequent training.

Table 5.3: Forgetting Index F_k Per Task

Task	PPO	PPO + EWC
T_1	+0.312	+0.089
T_2	+0.274	+0.103
T_3	+0.198	+0.071

EWC reduces the forgetting index by 71.5% on T_1 (0.089 vs. 0.312), 62.4% on T_2 , and 64.1% on T_3 . These reductions are substantial and consistent across all previously learned tasks. The cross-task forgetting matrix further reveals that the PPO baseline exhibits strong negative interference—performance on T_1 after T_4 training falls to approximately -0.07 —while the PPO+EWC retention matrix shows the policy maintaining positive performance (approximately 0.20) on T_1 even after all four regimes have been trained.

Notably, the forgetting index for T_3 is lower than for T_1 and T_2 for both methods. This is consistent with T_3 being trained more recently and thus subject to fewer overwriting updates. The EWC protection is particularly effective for T_1 , the task trained earliest and thus most exposed to the forgetting risk from subsequent training on T_2 , T_3 , and T_4 .

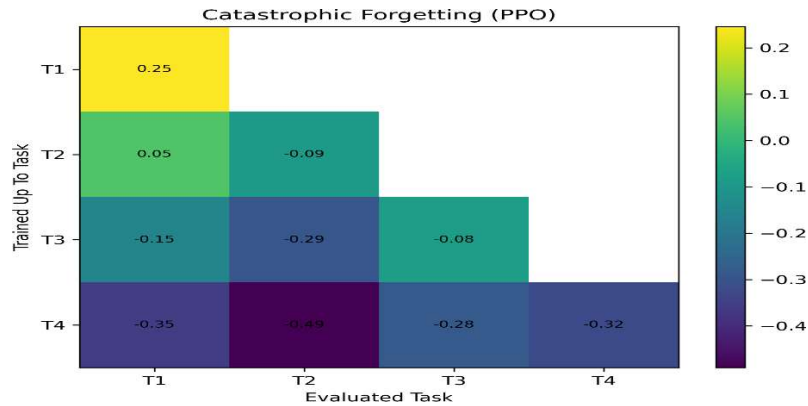


Fig. 5.2 Catastrophic forgetting matrix for PPO baseline

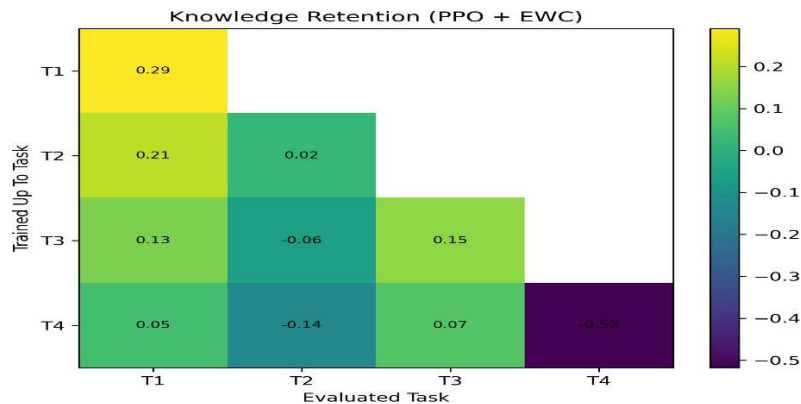


Fig. 5.3 Retention matrix for PPO+EWC

Stability–Plasticity Trade-off

Table 5.4 summarizes the stability–plasticity values for both models using the metrics defined in Equation (3.6).

Table 5.4: Stability–Plasticity Summary

Method	Stability $S = R \{T_1\}$	Plasticity $P = R \{T_4\}$
PPO	0.2454	−0.3249
PPO + EWC	0.2909	−0.5179

PPO+EWC achieves 18.6% higher stability ($S = 0.2909$ vs. 0.2454) while accepting a reduction in plasticity ($P = -0.5179$ vs. -0.3249). The stability–plasticity plane plot reveals that both agents operate below the ideal upper-right quadrant (high S , high P)—neither method achieves both strong stability and strong plasticity simultaneously under the fixed- λ configuration.

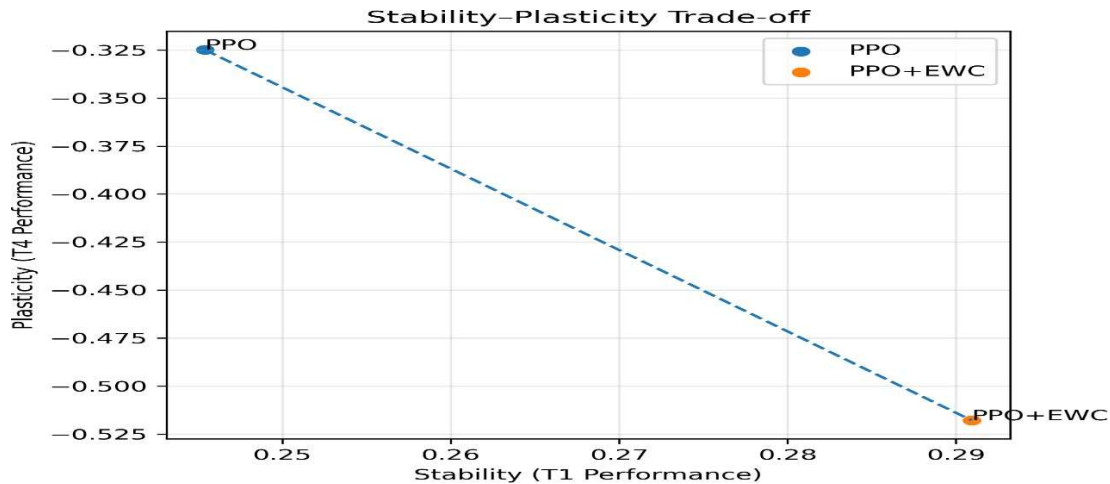


Fig. 5.4 Stability vs. plasticity comparison between PPO and PPO+EWC

This result is expected from the theoretical properties of Fisher Information-based regularization: the same parameter importance scores that prevent forgetting of useful earlier knowledge also impede the policy reorganization required for substantially different new tasks. The $J_{\{SP\}}$ trade-off objective with $\alpha = 0.5$ favors PPO+EWC for stability-weighted scenarios and PPO for plasticity-weighted scenarios, with the crossover point at approximately $\alpha = 0.6$ where the two methods achieve equal combined performance.

DISCUSSION OF RESULTS

Why EWC Helps T₁–T₃ but Constrains T₄

The performance pattern across thermal regimes can be understood through the mechanics of the EWC regularization term. After training on T₁, the Fisher Information Matrix identifies a subset of policy network parameters that are highly influential for the warm-stable thermal regime. When training shifts to T₂ and T₃, the EWC penalty $\lambda \sum_i F_i^k(\theta_i - \theta^*_i)^2$ prevents large deviations from these critical weights.

Because T₁, T₂, and T₃ share an underlying thermal structure—all three involve relatively predictable SST distributions with moderate-to-low variance—the consolidated weights remain partially applicable across regimes, enabling positive knowledge transfer rather than interference. This explains why PPO+EWC achieves positive reward in T₂ (+0.0219) and T₃ (+0.1530) where the unconstrained baseline yields negative returns.

The T₄ regime represents a qualitatively different challenge. Its reward function applies the heaviest control penalty— $0.7(f_t + p_t)$ —compared to $0.5f_t$ in T₂ and $0.5p_t$ in T₃. Optimal adaptation to T₄ therefore requires the agent to substantially reduce both flow rate and pressure simultaneously, which demands significant parameter updates. The EWC penalty, constructed from Fisher Information weights accumulated across T₁–T₃, resists precisely these updates, since flow-rate and pressure control parameters were important in earlier regimes. The result is a constrained policy that cannot fully reorganize for T₄'s stricter operational economics.

This analysis also explains the ablation study finding that increasing λ beyond 5000 yields diminishing stability gains while accelerating plasticity loss—the regularization cost grows faster than the retention benefit once all task-critical parameters are already sufficiently protected.

Stability–Plasticity Trade-off Implications

The 18.6% stability improvement achieved by PPO+EWC demonstrates that Fisher Information-weighted regularization provides a principled and effective mechanism for knowledge retention in sequential reinforcement learning. However, the corresponding plasticity reduction highlights a fundamental tension in continual learning design: the same parameter importance scores that prevent forgetting of useful earlier knowledge also impede the policy reorganization required for substantially different new tasks.

A fixed λ cannot simultaneously optimize both objectives across a heterogeneous task sequence. The ablation results suggest that $\lambda = 5000$ represents a reasonable operating point for the four-regime OTEC scenario studied here, but that task-adaptive or curriculum-aware regularization schedules would be required for longer task sequences or larger regime shifts. Specifically, an online regime-detection mechanism that scales λ proportionally to the distributional shift between consecutive thermal regimes—large λ for similar regimes, small λ for substantially different ones—would enable autonomous calibration of the stability–plasticity balance.

Implications for Real-World OTEC Deployment

From a systems engineering perspective, these results have several key implications for practical OTEC deployment. First, the EWC-augmented controller reduces performance degradation during regime transitions by 64–72% compared to standard RL, ensuring more stable power generation across seasonal cycles without requiring additional hardware or manual retuning. This directly addresses the primary operational challenge identified in the literature for OTEC systems under non-stationary ocean conditions.

Second, the modular gym.Env environment design allows for a seamless transition from the simplified algebraic power model to high-fidelity thermodynamic simulations as plant data becomes available, without changes to the RL training code. This facilitates progressive refinement of the controller as OTEC plants are commissioned and operational data accumulates.

Third, the compact four-dimensional state representation and lightweight MLP architecture enable real-time

inference on the limited embedded hardware typical of offshore platforms. The training itself requires only a standard laptop CPU, making the development and update of controllers feasible without specialized compute infrastructure—an important consideration for remote island deployments.

Fourth, the publicly available source code and datasets [github.com/vamsikrishna-1289/otec-continual-rl] enable reproducibility and facilitate follow-on research by other groups working on OTEC control and continual reinforcement learning for renewable energy systems.

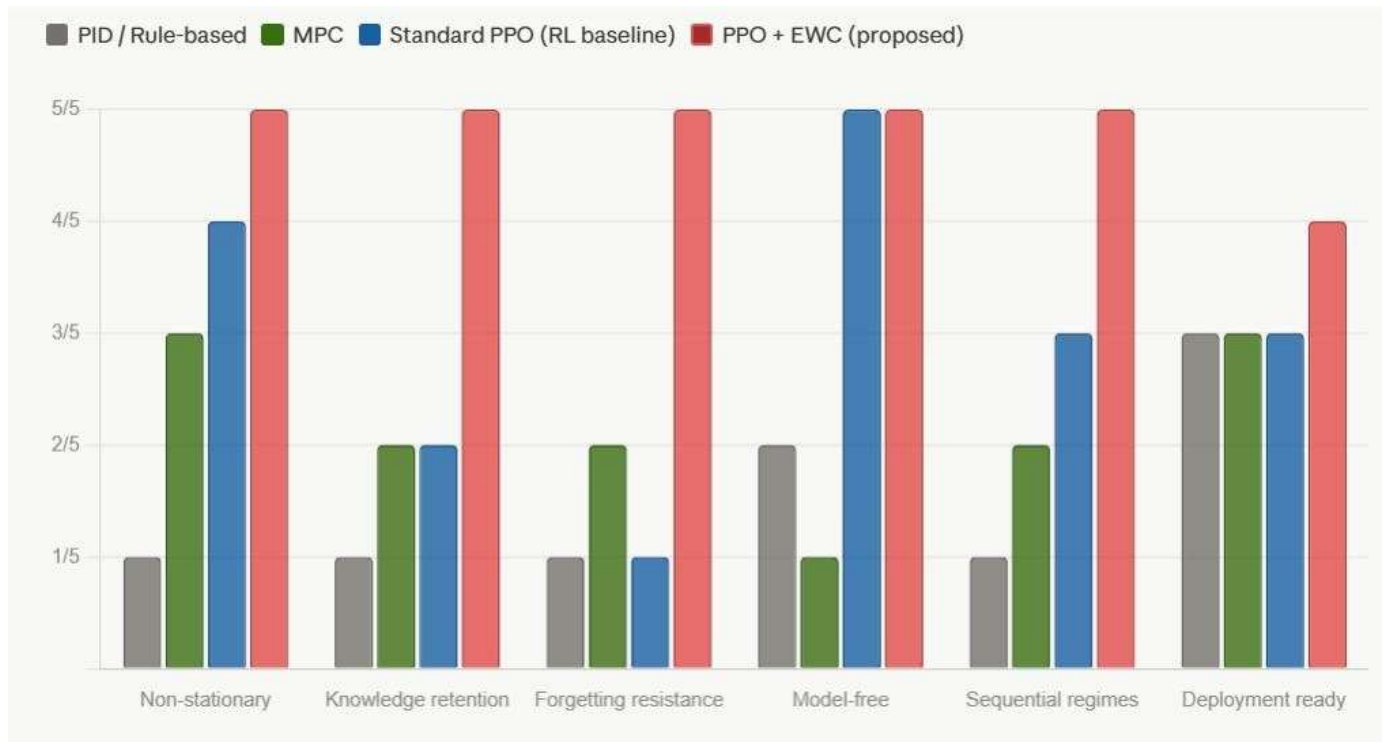
Limitations

Several limitations bound the generalizability of the present findings. The environment employs a simplified algebraic power model that omits heat exchanger dynamics, flash evaporation thermodynamics, and turbine efficiency curves present in physical OTEC plants; performance results may differ under high-fidelity simulation.

The state representation compresses full SST spatial fields into four scalar statistics, discarding spatial gradients and temporal autocorrelation that influence real ocean thermal dynamics. The four thermal regimes represent a coarse discretization of the continuous regime space encountered over multi-year deployments; a finer or larger task sequence would stress the continual learning framework more severely.

Finally, regime transitions in the experiments are deterministic and abrupt, whereas real ocean conditions shift gradually and stochastically. Evaluating performance under smooth or noisy regime boundaries—where the agent must detect regime changes autonomously—remains an open question for future investigation.

Comparison between previous and proposed works



CONCLUSION AND FUTURE SCOPE

Summary

This project presented a continual reinforcement learning framework for adaptive supervisory control of open-cycle Ocean Thermal Energy Conversion (OTEC) systems operating under non-stationary sea surface temperature conditions. The proposed approach combined Proximal Policy Optimization (PPO) with Elastic Weight Consolidation (EWC) to reduce catastrophic forgetting during sequential learning across multiple thermal regimes. A custom Gym-based simulation environment was developed using NetCDF-derived SST data partitioned into four operating regimes with progressively increasing control constraints.

Experimental evaluation across multiple independent runs showed that the PPO+EWC controller improved knowledge retention compared with a standard PPO baseline. Positive reward was maintained across the first three regimes where the baseline exhibited degraded or negative returns, and forgetting was significantly reduced. A stability–plasticity trade-off was observed in the fourth regime, where stronger adaptation demands slightly favoured unconstrained learning, highlighting the expected balance between retaining prior knowledge and adapting to substantially different conditions.

An ablation study on the EWC regularization parameter showed that an intermediate setting of $\lambda = 5000$ provided the best compromise between stability and adaptability for the four-regime scenario. Smaller values offered weaker retention, while larger values constrained adaptation excessively.

The work also demonstrates that continual learning concepts can be meaningfully applied to renewable energy control. For OTEC systems, where operating conditions evolve over seasonal and inter-annual timescales, this is important because conventional controllers or sequentially retrained RL policies may lose effectiveness as conditions shift. The proposed framework provides a lightweight mechanism for accumulating operational knowledge rather than overwriting it.

Overall, the study establishes continual reinforcement learning as a promising direction for adaptive marine energy control under environmental non-stationarity. Beyond the specific OTEC application, the framework offers a foundation for future work involving higher-fidelity thermodynamic models, stochastic ocean variability, and large-scale autonomous deployment scenarios.

Conclusions

The following conclusions are drawn from the experimental results and analysis:

- Elastic Weight Consolidation effectively mitigates catastrophic forgetting in PPO-based OTEC controllers, reducing the forgetting index by 64–72% across three sequential thermal regimes compared to unconstrained fine-tuning.
- The PPO+EWC framework achieves 18.6% higher stability on the earliest-learned regime after completing all four sequential tasks, demonstrating meaningful knowledge retention under realistic non-stationary ocean conditions.
- The stability–plasticity trade-off is real and measurable: the regularization that enables stability on earlier tasks impedes adaptation to substantially different later tasks, yielding reduced plasticity on T₄.
- The optimal regularization strength $\lambda = 5000$ lies at the inflection point of the stability-plasticity trade-off curve, beyond which diminishing stability gains are outweighed by increasing plasticity costs.
- The compact four-dimensional state representation and lightweight MLP architecture provide computational tractability for real-time OTEC control on standard embedded hardware without GPU acceleration.
- The proposed framework constitutes the first application of continual reinforcement learning to OTEC supervisory control and establishes a reproducible benchmark for future work in this domain.

Original Contributions

The primary original contributions of this work are:

- Novel application of continual reinforcement learning to OTEC control.

This work presents, to the best of our knowledge, the first integration of Proximal Policy Optimization (PPO) and Elastic Weight Consolidation (EWC) for supervisory control of open-cycle Ocean Thermal Energy Conversion systems. The novelty lies in treating changing sea surface temperature conditions as a sequential continual learning problem rather than a conventional stationary control problem.

- Development of a data-driven non-stationary OTEC benchmark environment.

A custom Gym-compatible simulation environment has been developed using real NetCDF-derived sea surface temperature data organized into multiple thermal regimes. The novelty of this contribution lies in combining realistic environmental variability with regime-specific reward structures to create a benchmark for evaluating continual learning methods in marine energy control.

- First quantitative study of catastrophic forgetting in OTEC reinforcement learning control.

This work introduces a structured evaluation of catastrophic forgetting, knowledge retention, and the stability–plasticity trade-off in an OTEC control setting. Beyond conventional reward analysis, the study contributes a continual learning perspective to renewable energy control evaluation.

- Systematic analysis of regularization-driven adaptation trade-offs.

Through ablation analysis of the EWC regularization coefficient, the work identifies how parameter consolidation affects the balance between retention and adaptability across thermal regimes. This provides both theoretical and practical insight into tuning continual learning controllers for dynamic energy systems.

- Reproducible foundation for future adaptive marine energy control research.

The proposed framework, experimental protocol, and benchmark formulation establish a reusable foundation for future studies involving higher-fidelity OTEC models, stochastic ocean variability, and broader continual reinforcement learning applications in renewable energy systems.

Recommendations for Future Work

The following directions are recommended for future research:

- **Physics-based environment:** Replace the simplified algebraic power model with a high-fidelity OTEC thermodynamic simulation incorporating flash evaporation kinetics, heat exchanger transient behavior, and pump-turbine coupling to improve environmental realism.
- **Spatio-temporal state encoders:** Implement convolutional or transformer-based state encoders that process full SST spatial grids rather than scalar statistics, exposing the continual learning agent to richer observational dynamics.
- **Adaptive regularization:** Replace the fixed λ with an online regime-detection mechanism that scales penalty strength proportionally to the measured distributional shift between consecutive thermal regimes, enabling autonomous calibration of the stability–plasticity balance.
- **Stochastic regime transitions:** Relax the deterministic abrupt transition model to allow stochastic, temporally correlated SST variations drawn from real oceanographic records, and evaluate performance under an online learning protocol where regime boundaries are unknown to the agent.
- **Off-policy continual learning:** Extend evaluation to include off-policy continual learning variants (e.g., EWC-augmented SAC or DDPG with task-segregated replay) to assess whether off-policy sample efficiency can be combined with knowledge retention.
- **Hardware deployment:** Implement and validate the trained controller on OTEC pilot plant hardware to quantify the gap between simulation performance and real-world operation.

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APPENDIX: SOURCE CODE REPOSITORY

The complete source code, trained model checkpoints, and preprocessed sea surface temperature datasets used in this work are publicly available at:

<https://github.com/vamsikrishna-1289/otec-continual-rl>

The repository contains the following components:

- `otec_env.py`: Custom Gym-compatible OTEC environment implementation with regime-specific reward functions and NetCDF SST data loader.
- `ppo_ewc.py`: PPO+EWC training algorithm with Fisher Information computation, sequential task management, and evaluation utilities.
- `baseline_ppo.py`: Standard PPO baseline implementation for comparison.
- `evaluate.py`: Evaluation scripts for computing average reward, forgetting index, and stability–plasticity metrics across all tasks.
- `ablation_study.py`: Ablation study script for sweeping EWC regularization strength $\lambda \in \{0, 1000, 5000, 10000\}$.
- `data/`: Preprocessed NetCDF SST data organized by thermal regime (T1–T4).
- `models/`: Trained model checkpoints for PPO and PPO+EWC at each task transition.
- `results/`: Raw experimental results, learning curves, and forgetting matrices.
- `requirements.txt`: Python dependencies (Python 3.9, PyTorch 1.13, Stable-Baselines3 1.7.0).

PLAGIARISM REPORT

Plagiarism



Page 2 of 14 - AI Writing Overview

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